

Cynthia Cesar

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OBJECTIVE

Electrical and Computer Engineering student with an A.A. in Electrical Engineering, seeking an Engineering Internship where I can apply hands-on skills in embedded systems, robotics, digital logic, and prototyping. I take projects from breadboard to soldered build, from Arduino and Raspberry Pi systems to 3D-printed mechanisms, and I am motivated by building practical technology through iterative testing and thoughtful engineering.

EDUCATION

Miami Dade College, Kendall Campus — Kendall, FL

Completed June 2025

Associate of Arts in Electrical Engineering

- **Relevant Coursework:** Circuit Analysis (Direct Current), Digital Logic
- **Research:** Introductory Epoxy Metal Polishing Research

TECHNICAL SKILLS

- **Programming:** Python, C++, Arduino (Embedded Programming), Raspberry Pi
- **Digital & Electrical Engineering:** Circuit Analysis (DC), Digital Logic, BCD Counters, Combinational Reset Logic, PWM Motor Control
- **Electronics & Hardware:** Circuit Prototyping (Breadboard to Perf-Board), Soldering, Multimeter, Oscilloscope, DC Power Supply, Hardware Debugging, Buehler EcoMet 30
- **CAD & Prototyping:** SolidWorks (Proficient), Fusion 360 (Intermediate), 3D Printing

ENGINEERING PROJECTS

Smart Jack-in-the-Box — CircuitBox Audio Amplification Systems

Sept 2025 – Oct 2025

- Built an Arduino-driven interactive jack-in-the-box with a DS3235 35 kg servo, ultrasonic sensor, LED signaling, and custom audio amplification.
- Designed and soldered the complete wiring schematic, tested multiple speaker configurations to improve audio output, and debugged electrical connections.
- Constructed a magnetically retracting wooden enclosure that coordinates the box's movement with the electronic control system.

Biomimetic Vision & 3D Tracking System

Aug 2025 – Dec 2025

- Engineered a modular animatronic eye system on a Raspberry Pi 4 with real-time computer vision to detect, track, and follow individuals indoors.
- Designed and 3D-printed a pan-tilt eye assembly with MG90S servo motors, replicating human-like gaze and synchronized eyelid motion.
- Separated sensing from actuation: the Raspberry Pi handles vision while a microcontroller executes precise PWM-based motor control.

BCD 24-Hour Digital Clock

March 2026 – May 2026

- Independently designed and built a fully synchronous 24-hour BCD clock (00:00:00–23:59:59) using a 32.768 kHz crystal oscillator, CD4060, and cascaded CD4518 counters.
- Integrated combinational reset logic for 60-second, 60-minute, and 24-hour rollover, plus a gated time-set control for manual adjustment.
- Advanced from breadboard prototyping to a soldered perf-board build, with PCB fabrication and enclosure design planned.

FPV Drone

2026 – Present

- Building a drone from the ground up on a custom carbon fiber frame with brushless motors, ESCs, flight controller, radio receiver, FPV camera, and power distribution.
- Soldering, wiring, configuring the flight controller, and tuning flight through iterative testing and troubleshooting.

TECHNICAL & PROFESSIONAL EXPERIENCE

Robotics & Coding Instructor — Bricks 4 Kidz | Tekniklabs, Kendall, FL

May 2026 – August 2026

- Designed and delivered hands-on STEM lessons in robotics, programming, and electronics. Adapted instruction to different skill levels and encouraged students to troubleshoot failed designs, test solutions, and iterate.

Retail & Shipping Associate — FedEx Office, Doral, FL

November 2025 – May 2026

- Processed domestic and international shipments, helped customers troubleshoot printers and other equipment, and completed print production orders while managing high-volume periods accurately.

Summer Shade Guide / Camp Counselor — Branches, Inc., Florida City, FL

Summer 2024 – Summer 2025

- Supervised kindergarten and first-grade students and planned and led educational lessons, creative activities, and group games.

LEADERSHIP & INVOLVEMENT

- Vice President, MDC Kendall Society of Women Engineers (SWE) Chapter
- IEEE Member (Aug 2025 – Present): support faculty-led research in precision polishing and materials finishing, including materials characterization and component testing
- Organic Gardening Volunteer, Miami Dade College, Kendall Campus (2026 – Present)